

EXECUTIVE BRIEFING

Start with the bank. Keep the decision checkable.

ZQUAS governs decisions made by automated systems and AI agents. Financial crime is where it starts.

TRL 6. Synthetic data only. No production deployments.

What the bank evaluates

A 12-week local pilot on its own data and infrastructure. An optional network phase follows by separate agreement, after the local results have been reviewed.

Local monitoring and the network

The local monitor identifies movement patterns for an analyst to investigate. Its full-world evaluation has not yet been measured. The separate network work lets participating institutions add bounded signals to their own decisions. Network signals add reviews, never automatic holds.

Records and sharing

No customer records leave your infrastructure. With network sharing switched on, the bank sends only bounded answers about payments it is part of, each one authorized by its own policy and recorded in its signed log.

Signed decisions

Every decision is signed, and the supervisor can verify it with a published check (synthetic data, development keys).

Programmes and sandboxes

ZQUAS is taking part in the FCA's Supercharged Sandbox. The FCA does not endorse ZQUAS, its products or its results.

ZQUAS was accepted into the FCA's Digital Sandbox in March 2026. The FCA does not endorse ZQUAS, its products or its results.

ENGINE EVIDENCE

Four figures. The scope of each.

Internal tests on synthetic fixtures. Each figure has its own workload and measurement boundary. These are not full product performance guarantees or an external audit.

500K entities under 2 seconds

500K entities under 2 seconds on the fixed synthetic fixture, with about 55K transactions. This is a detection test, not the full product cycle.

alert lifecycle under 10 ms

Alert lifecycle under 10 ms per alert, from a detection result to a persisted, queryable alert. Batch results are amortised per alert.

federation bilateral round under 10 seconds

Federation bilateral round under 10 seconds on TCP loopback development transport. Semi-honest security. This does not measure a real inter-bank network.

VRAM under 1 GB

VRAM under 1 GB in the measured detection-pass working set after warm-up. Resident ingestion and model inference are excluded. This is not total-system memory.

Read the boundaries together

The timing fixture has about 55K transactions. The memory fixture is a separate detection-pass measurement. Neither proves full-system performance at a realistic transaction density with all components resident. The memory regression gate is 1 GiB; the recorded detection-pass working set is below 1 GB.

Methods and limitations: zquas.ai/benchmark.html

THE PILOT

12 weeks locally. A network phase by agreement.

Who participates

A bank or payment firm exploring local transaction monitoring, with a compliance owner, an analyst or operations lead, and an IT contact.

What the bank supplies

An agreed local dataset and its permitted use, account and payment history, and the outcomes available for evaluation. Suitable infrastructure, including an NVIDIA GPU. The bank completes its own security, legal and procurement checks.

What ZQUAS supplies

Pilot software, support to configure the local monitor, the signed decision log and its verification check. Setup, evaluation support and the report are agreed before the pilot starts.

What gets measured

Scam payments caught, analyst workload, and whether every decision can be verified in the signed log. Where payment review and hold policies are evaluated, the limits are at most 1 review per 500 payments and 1 hold per 1,000. These are evaluation limits, not promised detection rates. Local transaction monitoring itself selects investigation work and does not hold payments.

What happens afterwards

Review the evidence, limitations and operational fit. Decide whether to continue locally, change scope, or stop. An optional network phase requires a separate agreement on participants, sharing policy, security, legal basis and success measures.

The local phase starts after scope, agreements, access and data readiness are confirmed. Discuss the pilot: zquas.ai/founding-partner.html

LOCAL MONITORING

One synthetic walkthrough.

All people, accounts and payments here are fictional. The monitor has not yet been measured on the full synthetic worlds. No detection rates or alert counts are claimed.

Several payers in | Synthetic example

Several synthetic payers send money into one account. The bank uses the incoming payments in its own journal.

Money out within a day | Synthetic example

Money is debited out within a day of arriving. An outgoing debit is not proof that another bank received it.

The pattern score | Synthetic example

The local monitor scores a funnel or rapid pass-through pattern using its own observations and frozen calibration. With sufficient history, it combines movement with context from earlier activity. The score can nominate work for review. It does not establish that a crime occurred.

The analyst queue | Synthetic example

Nominations are ranked. Admission depends on the daily analyst-time limit and monthly investigation allowance. This local monitoring step does not hold a payment.

The signed tm-alert line | Synthetic example

The monitor writes a tm-alert line for a new queue entry, admission or expiry. Review and refusal decisions carry a proof, and the institution signs the log head. The supervisor can check the record using the public key. Synthetic data, development keys.

The bank decides | Synthetic example

The analyst examines the bank's own evidence and follows its procedures. The bank decides what action is justified. The pattern alone does not establish wrongdoing.

Read the walkthrough: zquas.ai/use-case.html

What is established. What remains to measure.

Internal evidence

The engine figures were checked against the F1_ENGINE code, claim register and retained reports. The fixed 500K detection fixture, per-alert handling, loopback federation and detection-pass memory each have a different scope. They must not be combined into a full-system performance promise.

Signed records

The published supervisor check verified the synthetic logs in run R43.0 B, using development keys. The local monitor also records its queue decisions as tm-alert lines under signed log heads. Verification establishes the integrity of that record, not the correctness of a criminal allegation.

Local monitor

The walkthrough follows kit tm-run, its frozen feature sheet and the current queue and logging implementation. Full-world detection rates and alert counts are not yet measured. This briefing does not publish them.

Network phase

Current federation security is semi-honest: participants are assumed to follow the protocol. Deliberate protocol deviation is outside that security claim. The optional network phase needs separately agreed sharing rules, legal review and operational measures.

Further reading

zquas.ai/benchmark.html

zquas.ai/use-case.html

zquas.ai/founding-partner.html

zquas.ai/contact.html

ZQUAS B.V., the Netherlands. This is a pilot discussion document. Terms and scope are subject to agreement.